

began to be digitised. Eventually, the digitised records reached a critical mass and have now become the primary way of storing an individual's healthcare information. Pharmaceutical companies now involve data scientists to process those immense volumes of patient data for various objectives – patterns related to treatments, side-effects of drugs, and cost-benefit analyses are all emerging from the rich minefield of big data.

The nature of healthcare data is inherently extremely complex. It includes data of a scientific, demographic, personal data and, to an extent, qualitative nature as well. This complexity and richness of the data has prompted several software companies to develop analytical tools built specifically for the healthcare industry. Such companies and their tools have proliferated in the past half a decade or so.

Over the past few years, there have been several factors that have collectively given big data the requisite push to go mainstream in the healthcare industry as well.

First, increasing healthcare costs have brought in a need for a change in the approach to healthcare. Also, with the incomes of pharmaceutical companies as well as physicians being increasingly linked to the success of the treatment and not just the treatment numbers alone, there is now added incentive to find effective treatments. Second, the advent of dedicated healthcare analytics tools, along with the availability of large scale data, has made it feasible to apply big data in finding treatments. Finally, seeing how big data has brought positive results in other fields, it would appear to be a good bet for the healthcare domain as well.

Indeed, remarkable improvements have been made in healthcare once the medical records were available and the data could be mined for patterns. It has set up virtuous cycles that sustain themselves.

First, the trends in medical data have led to people being able to make cleaner lifestyle choices on their own, allowing people to take responsibility for their own well-being. Prevention is better than cure. The advent of wearable tech has allowed for fitness tracking devices such as the Fitbit and Samsung Gear Fit to become commonplace. The data from your body can be compared in real time with millions of people around you to predict an illness accurately, before it even occurs.

Second, treatments are now dispensed with the added support of empirical evidence. Treatments are now described not only in terms of biological impact, but also include a probabilistic description of their effectiveness. This kind of mathematical confidence has given a great boost to the credibility